Lane — Master Features, Design, and Backend Plan

> **Planning and consolidation only.** This document records the intended Lane product, design direction, data boundaries, and staged delivery plan from the supplied source-of-truth, area reports, dated audit summary, and latest owner feedback. It is **not proof that any feature is built, connected, tested, legally approved, or ready to submit**.**No award guarantee.** Lane can be a focused, meaningful Congressional App Challenge project, but this plan cannot promise an award, a ranking, no defects, a passing permit score, or real-world driving readiness.
> **Document status:** Master consolidation for future implementation and review
> **Product:** Lane, a phone-first mobile-web/PWA learning experience for teens preparing for California permit knowledge and early driving practice
> **Latest product authority supplied:** Owner feedback dated October 12, 2026, interpreted alongside the agreed source of truth
> **Repository caveat:** The actual latest VS Code repository and runtime are **not accessible in this planning task**. The supplied audit is dated/snapshot evidence, not current verification.

Quick first-page summary

Decision	Recommended direction	What must not be claimed yet
Product shape	A calm, phone-first learning feed that borrows familiar social-media patterns for useful driving and permit learning.	That Lane is an endless social network, official DMV instruction, or a replacement for driver education/supervision.
Home	Logo top-left; compact streak/lessons crossbar; subtle XP; real Continue Learning; Daily Goal; honest local/demo leaderboard; no filler below.	That the visual fixes or a real global leaderboard are already live.
Core navigation	Persistent bottom navigation on Home, Feed, Practice, Progress, and Profile; back/exit only for genuine drilled-in views, settings, and simulators.	That Feed already has this consistency in the latest code.

Feed	Direct local HTML video plus permit posts, flashcards, glossary, visuals, focused quizzes, and relevant practice. Media stays dominant.	That all 19 videos play on devices, have captions, or are offline-cached.
Learning checks	Checks follow the triggering item and provide clear choices: answer, key segment, help, or Review Later. No random trap or scroll-lock design.	That current quiz dismissal, review queue, or correction flow is fixed.
Practice and Progress	Preserve the liked Practice page and useful Road Ahead/Progress information; refine only concrete issues and reduce overly game-like styling.	That current practice results feed the learning algorithm or that readiness is validated.
Simulations	Exactly 10 short post-quiz scenarios: 7 animated 2D and 3 driver-seat 3D. GPT/Codex owns package refinement; Claude/Claude Code owns app integration.	That the ZIP packages are integrated, handbook-approved, or behaviorally verified.
Personalization	Explainable evidence, not popularity or time-spent ranking: Accuracy, Work, Attention, and Time quality.	That a backend, per-topic mastery, normalized Work policy, or remote sync is implemented.
Backend path	Local-first behind the existing progress seam; connect the approved remote service only after its configuration, access rules, migrations, and end-to-end save/read are verified. Supabase is a documented candidate, not a verified requirement.	That browser persistence proves cloud sync, auth, public comments, or real leaderboards.
Teen safety	Supportive correction, privacy-minimizing profile, nickname-only future public display, no fake comments, and no use while driving.	That Privacy Policy/Terms, moderation, or legal review is complete.

Recommended product thesis

Lane should be **the useful alternative to unstructured scrolling**: a learner opens to an actual next driving lesson or review, sees familiar media controls without fake social claims, gets a short check at an understandable moment, and can defer a topic honestly without losing it. The experience should feel modern and encouraging, not punitive, addictive, overly childish, or like a large driving game.

Implementation principle

> **Preserve the good existing app before adding breadth.** Keep the established Lane visual language, Practice page, Road Ahead/Progress information, local progress seam, routes, and assets. Improve the proven gaps in small tested slices. Do not replace the app architecture or invent a new frontend just to add a feature.

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1. Purpose and latest product direction

1.1 What Lane is for

Lane is a **supplemental educational mobile-web experience** for teens who are planning to drive, studying for a California permit test, or beginning supervised driving practice. It combines practical driving lessons with California permit knowledge in a short-form, phone-friendly feed.
It is not a replacement for:

- certified driver education;
- adult supervision;
- real behind-the-wheel practice;
- California DMV guidance or an official DMV product;
- emergency services; or
- a promise that a learner will pass a permit test or drive safely in every situation.

1.2 Latest recommended design direction

The latest owner feedback should guide the next design pass where it revises older presentation decisions:

1. **Simplify Home rather than rebuild it.** Show the real Lane logo at upper-left, remove unexplained centered header text and the prominent Home Saved button, keep Saved capability elsewhere, and stop the page after useful content.
2. **Make learning progress quiet but meaningful.** Use one compact streak/lessons crossbar and put total XP underneath as supporting information.
3. **Make Continue Learning honest.** It must open the learner’ s actual next unfinished, current, or due review item rather than decorative filler.
4. **Retain useful motivation without pretending there is a social service.** Keep Daily Goal and the leaderboard below it. Local/demo ranking must never masquerade as a live public ranking.
5. **Keep a stable feed shell.** Persistent bottom navigation belongs on ordinary core screens. A back arrow belongs only on a true detail view, settings flow, or full-screen/in-feed simulation exit.
6. **Use familiar feed controls carefully.** Heart/like, comment affordance, share, save/bookmark, and separate sound control can make content recognizable; none may imply fake comments, public likes, or a cloud social graph.
7. **Repair placement, not the learning safeguard.** The owner dislikes awkward, apparently random Quick Recall/Quick Safety Check interruptions. The correct response is clear placement and visible deferral—not removing important learning checks and not locking scrolling as punishment.
8. **Preserve what works.** Keep the liked Practice page; retain useful Progress/Road Ahead and Profile content; soften overly Duolingo-like presentation without structural redesign.
9. **Make simulations feel intentional.** They need a purposeful pre-start state, active slow motion rather than frozen traffic, clear choices, non-graphic cartoon impact/warning

symbols for unsafe outcomes, and correct results that advance.

10. **Keep scope ownership clear.** GPT/Codex refines simulation packages. Claude/Claude Code integrates approved packages and handles the rest of Lane’ s frontend, content flow, learner state, and future backend seam.

1.3 Design values for teen users

Value	Practical application	Avoid
Respect	“Almost. Here is the rule that changes the answer.”	Failure labels, lockouts, shame, or treating skipped help as a character flaw.
Clarity	State why a quiz, review, or practice activity appears.	Surprise traps, unexplained score changes, mysterious rank, or decorative progress claims.
Familiarity	Use recognizable feed controls and a consistent bottom navigation bar.	Copying social-media engagement incentives or obscuring media with control clutter.
Calm attention	Short sessions, one focused question, optional/need-based practice.	Infinite-scroll objectives, streak punishment, urgency pressure, or a speed-game tone.
Safety	California/US terminology, handbook-backed learning items, no-use-while-driving message.	Crash spectacle, medical/legal assertions, unsafe emergency guidance, or certification claims.
Honesty	Label device-only/demo behavior and unavailable features accurately.	Fake public comments, fake friends/rankings, false sync success, or “AI knows you” language.
Accessibility	Captions/text alternatives, visible focus, reduced motion, touch targets, safe areas.	Requiring sound, hover, fast reactions, or unannounced motion to learn a rule.

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2. Source confidence and status labels

2.1 How to read this document

This document intentionally separates product intent from build evidence. The supplied materials include historical plans, an agreed source-of-truth, a supplied audit report, and the owner’ s later hands-on feedback. They do not give this task live access to the current repository or device runtime.

Label	Meaning in this plan	It does **not** mean
OWNER-AGREED	An explicit feature, behavior, or product boundary in the source-of-truth or confirmed owner direction.	The feature is implemented or has passed tests.
LATEST OWNER DIRECTION	A later owner preference or issue observed while using the item; it overrides older visual intent where it conflicts.	A completion report or a technical diagnosis.
AUDIT-REPORTED	A supplied audit described a code/runtime observation in its audited snapshot.	That the latest VS Code repository or a physical device matches it today.
HISTORICAL	Earlier discussion, starter implementation, rough concept, or superseded method.	A current requirement, current backend, or current implementation.
DOCUMENTED / PLANNED	A design or backend specification exists in the supplied documents.	A configured remote service, migration, or end-to-end feature.
UNVERIFIED	No reliable evidence in this task proves current behavior, device behavior, data persistence, or owner approval.	That the feature is impossible or should be deleted.
DECISION NEEDED	A material policy or integration conflict must be selected explicitly before implementation.	Permission for an implementer to choose silently.
NEW / NOT YET APPROVED	A small recommendation in Section 9, intentionally separated from agreed scope.	An owner requirement or a feature to ship automatically.

2.2 Authority order when sources disagree

1. **Current, directly inspected source code and tested runtime** should decide what exists once a future implementation task has repository access.
2. **Latest owner direction** decides desired visual/product correction, especially the October 12 feedback.
3. **Agreed source-of-truth** defines the documented product baseline and safety boundaries.
4. **Supplied audit-reported evidence** describes the audited snapshot, including its strengths and gaps.
5. **Historical starter/backend materials** provide context only. They must not be copied into the actual app or treated as proof that its backend exists.
6. **Assistant suggestions** remain suggestions unless explicitly approved.

2.3 Important confidence cautions

- The supplied audit reported a local/demo Vite + plain TypeScript PWA and no Supabase client, API, SQL, or server in that repository snapshot. That does **not** prove the next checkout has the same state.
- The source-of-truth describes an earlier starter backend as historical context. It must not be confused with the audited Lane repository.
- The audit’ s reported build and 24 passing tests were evidence of that audit run only. They are not a current build, a backend connection, a live deployment, or release readiness.
- The owner’ s feedback identifies observed UX issues and desired corrections. It does not certify fixes, captions, licenses, content accuracy, hardware support, or simulation package behavior.
- The 19 videos are owner-confirmed Creative Commons-vetted for internal integration. Source, creator, attribution, caption status, and final public-release verification remain content workflow fields rather than claims of runtime completion.

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3. Chronological decision history

This history preserves the useful evolution of the design. It prevents old methods from being accidentally reintroduced as current requirements.

Period / iteration	Decision or method	Current reading and preservation rule

Early concept	Lane was framed as a healthier social-media alternative for teen driver learning.	Keep the useful feed metaphor; reject popularity, virality, and endless-scroll optimization.
September content/presentation work	The original assignment was to verify the supplied videos, produce a simple lesson/description/impact/quiz slideshow, download or capture and re-upload approved videos, and suggest missing topics. An overly complex deck was rejected and simplification requested.	These are curriculum/asset deliverables, not proof of app features. Preserve useful content and credits; never use the rejected elaborate presentation or old Manus frontend as the app design. “Sept. 19” was a retrospective status note, not a verified completion date.
Initial content planning	Earlier planning discussed a larger library, YouTube/Creative Commons discovery, Google Drive, and possible embeds.	Current runtime direction is **19 ordered local assets** with direct HTML video. Drive remains archive/source storage, not the in-app player.
Initial feed idea	Videos anchored the experience and quizzes appeared after lessons.	Retain the lesson → focused quiz → useful follow-up → relevant practice method, but avoid a video-only stream.
Mixed-content refinement	Flashcards, glossary, permit-test posts, posters/photos/artifacts, 2D activities, and selected 3D scenes were added.	Keep a mixed learning feed so the 19 videos remain useful over time. Treat authored permit posts as curriculum, not user posts.
Early quick-time concept	A scene/video might pause at a decision point and branch, then rewind/retry.	Historical only. Current simulation direction is active **slow motion/time dilation**, not a frozen scene.
Simulation scope refinement	Counts varied in earlier material: 10+, 12, 5 2D/4 3D, and a hard-coded yellow-light concept.	Locked target: **10 total: 7 animated 2D QTE scenes + 3 driver-seat 3D scenes.** Use the exact catalog in Section 6.
Simulation package boundary	Separate packages and manifests were proposed.	Preserve package ownership: GPT/Codex refines simulation packages; Claude/Claude Code imports and connects approved packages without redesigning them.

Unsafe-outcome feedback	Owner observed missing collision symbols and weak outcomes.	Latest correction allows non-graphic cartoon spiky impact/warning bursts, near-misses, blocked paths, and braking/conflict warnings; do not ban all depiction and do not use gore/injury.
Home presentation snapshot	Separate streak, XP, and lesson indicators, header text, Saved button, extra Home blocks.	Latest owner direction simplifies this into logo, compact crossbar, subtle XP, Continue, Daily Goal, leaderboard, then stop.
Navigation snapshot	Bottom navigation was reported on several screens, while Feed used a back arrow without the persistent bar.	Resolve the inconsistent shell: persistent bottom navigation on core feed screens; back only for truly drilled-in contexts.
Feed action snapshot	Action controls existed or were discussed but appeared too circular/overlaid; blog language appeared.	Preserve distinct social-style icons: heart, comment affordance, share, save, separate sound. “Blog” must not be conflated with public comments.
Quiz interruption snapshot	Quick Recall / Quick Safety Check cards felt random; current audit reported silent dismiss paths and same-question retry behavior.	Keep meaningful checks but attach them to the source item, show a reason, offer explicit deferral/help, and provide similar—not identical—correction questions.
Practice / Progress feedback	Owner liked Practice and much of Progress/Profile, but disliked a Duolingo-like feel.	Preserve their structure and useful content. Make small visual corrections only; do not replace the areas with a generic dashboard.
Profile refinement	Avatar/gamification was present in an audit snapshot; owner requested editable name.	Keep avatar simple, add a privacy-safe editable display name to the same learner record, and avoid an economy expansion.

Leaderboard decision	Some planning would remove social features; owner explicitly found leaderboard useful.	Keep its placement and local/demo value. Do not imply live peers, global rank, or public leaderboards without approved remote safety work.
Progress identity method	Local storage and a proposed anonymous learner identity appeared in different materials.	One learner record must unite profile, avatar, progress, XP, goals, review state, and permit date. Local persistence remains valid without a cloud service.
Backend assumption correction	Starter materials mentioned Supabase/Auth/Postgres/RLS; audit found none in the Lane repository.	Do not migrate, guess configuration, or claim cloud service. Add any remote service only behind an adapter after real discovery.
Framework correction	Historical sample stacks can suggest React, tRPC, Drizzle, Express, or a starter project.	Rejected for this app unless the owner explicitly changes scope. Preserve the established Vite + plain TypeScript PWA and its routes/design.
Learning signal refinement	The formula was intentionally simplified into Accuracy, Work, Attention, and Time quality.	Keep explainability. Time is a small diagnostic, not a speed contest; mastery and review priority are different decisions.
Work-score conflict	Written fixed Work weights and reference-code availability normalization differ; an assistant changed the missing-signal scoring rule without owner approval.	**DECISION NEEDED.** Preserve the conflict and current reference behavior until the owner selects a policy with tests. Do not silently alter weights or normalization.
Permit-date refinement	A test date was introduced as a way to adapt content.	Keep it optional and change only content balance. It cannot change the formula, erase progress, pressure the learner, or guarantee passing.
Content and legal workflow clarification	Captions, source/licenses, handbook validation, glossary rewriting, Novato/local-impact evidence, Privacy/TOS were discussed.	Preserve these as editorial/release workflows. They are not evidence of a backend or current runtime dependency.

3.1 Non-negotiable preserved methods

- Use **stable IDs**, not filenames, route positions, or display titles, to connect content, progress, quizzes, review items, and scenarios.
- Use **direct local/static media paths** at runtime; keep Drive/archive storage separate from playback.
- Use **purposeful learning stops**, not random popovers or invisible dismissal.
- Use **local-first persistence through one seam**; components should not scatter direct storage or future service calls.
- Use **content review records** for handbook source, credit/license, captions, and approval state rather than baking editorial facts into UI code.
- Use **small vertical slices** that prove a real learner action end-to-end before expanding the library.

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4. Recommended final screen-by-screen experience

4.1 Shared shell and visual rules

- **Top identity:** Lane logo at top-left; compact page context only when useful; no unexplained centered header text.
- **Bottom navigation:** consistent icons and labels for **Home**, **Feed**, **Practice**, **Progress**, and **Profile**. Existing route naming may differ, but the labels must be clear and stable.
- **Back/exit rule:** a back arrow is reserved for a true drilled-in detail, settings subpage, full-screen/in-feed simulation, or other temporary focus mode. It must return to the previous relevant position, not bounce unpredictably.
- **Feed width:** phone-first at 360×800 and 390×844; a readable centered column at tablet/desktop rather than stretched media cards.
- **Action controls:** effective 44px minimum touch/focus target, but light visual weight. Every icon has an accessible name, visible keyboard focus, and state where applicable.
- **Motion:** respect reduced-motion; no educational outcome depends on seeing fast animation alone; no timer silently runs while the tab is hidden.
- **Tone:** supportive, specific, and never punitive.

4.2 Home

Recommended order

```
Plain Text
Lane logo / quiet header
Compact crossbar: streak day | lessons completed
Subtle line: total XP earned
Continue Learning
Daily Goal
Leaderboard (honest local/demo presentation until a real service exists)
Bottom navigation
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Home element	Recommended visible behavior	Dependency and status
Logo	Real Lane logo in the upper-left; accessible link/label if it navigates.	OWNER-AGREED visual correction; not verified as implemented.
Compact crossbar	Current streak on the left, lessons completed on the right, total XP below. Values remain readable at narrow widths.	Uses local progress metrics first; presentation change remains unverified.
Continue Learning	One primary action whose label reflects actual stored next/current/due state. Include a clear empty/fresh state.	Basic next-unfinished selection was audit-reported; due/review intelligence is documented, not proven.
Daily Goal	Calm progress below Continue. Qualifying work—not scrolling or replay farming—advances it.	Local demo behavior was audit-reported; no sync claim.
Leaderboard	Below Daily Goal. Local mode must say “Your device progress” /equivalent if it lacks real rankings.	Owner wants it retained; public ranking is future/off.
Saved access	Removed from the Home hero/header but preserved in one discoverable library/Profile destination.	Capability must not be deleted merely because the header control moves.
Home ending	No “Make your next move” or redundant filler after leaderboard.	OWNER-AGREED removal; verify visual bottom state.

4.3 Feed

The Feed is the main learning surface, not an addictive infinite stream.

1. Open a relevant **direct video** for a fresh or returning learner.
    - The agreed default session entry is the Feed/video, not a mandatory Home dashboard first. Home remains available through navigation. A new learner gets Lesson 1 then Lesson 2; a returning learner gets a relevant next lesson/refresher. The logo, Home cleanup, and gamification changes do not override this opening behavior.
2. Keep media/content visually dominant and the bottom navigation present.
3. Add a small, honest action rail: heart/like, comment affordance, share, save/bookmark, and separate sound/mute.
4. Follow the source item with **one focused check** only when configured and explain why it appears.
5. Offer an immediately visible path to answer, watch the key segment, get help, or Review Later when the check is important.
6. Return a useful next item: flashcard, glossary, authored permit post, visual reference, next lesson, or related practice.
7. Do not display fake comments, fabricated likes, public activity, or a fictitious global rank.
   Feed action rules
Control	Honest behavior now / later
Like / heart	May be local/private preference if retained. It must not influence mastery, XP, streaks, review ranking, or public popularity.
Comment icon	A UI direction replacing the blog affordance. Until a moderated community exists, omit it or give an honest not-open state; never seed fake comments.
Share	Use native share or copy a valid link when technically possible. Do not show a fake “shared” success.
Save / bookmark	Prefer a clear mapping into Review Later or keep a clearly distinct Saved library. Do not leave two confusing duplicate queues.
Sound / mute	Separate from engagement actions. Audio cannot be the only way to learn a rule; captions/text status must be honest.

4.4 Practice

The owner likes the Practice page. Preserve its current recognizable learning/test flow and avoid a cosmetic replacement.

- Keep the reported Practice/permit-test experience, including the **Practice/46 audit snapshot** as a reference scope rather than proof that a current 46-question bank is available today.

- Keep full-row answer targets, visible selected states, keyboard access, and bottom-safe scrolling.
- Repair concrete issues only: target-size shortfalls, navigation overlap, source accuracy, useful result explanation, and responsive behavior.
- Do not convert every Practice attempt into public competition or automatic mastery.
- If future results become evidence, record topic-level knowledge carefully rather than treating one total score as overall readiness.

4.5 Progress / Road Ahead

- Keep useful roadmap, topic, and permit-preparation information the owner likes.
- Remove overly Duolingo-like visual excess without deleting structure or relabeling completion percentage as validated mastery.
- Show **written permit knowledge** and **practical decision practice** separately when enough evidence exists.
- Explain uncertainty with phrases such as “More practice can help” or “Limited evidence,” not a certificate or guarantee.
- Leave detailed formulas out of the main screen; link to concise “Why am I seeing this?” explanations.

4.6 Profile and Settings

Area	Recommended behavior
Profile	Simple avatar, editable display name, and a clear note if progress is saved only on this device. No forced account creation.
Editable name	Persist with avatar and learning progress in the same learner record. Validate empty/overlong/unsafe input and use nickname-only display if any future leaderboard is public.
Avatar	Preserve selection and any safe existing local behavior. Do not add trading, collections, extra currencies, or a social economy.
Permit-test date	Optional setting with edit/clear action and a calm explanation that it changes content mix, not pass odds.

Review/reminder preference	Reserve a setting for future opt-in reminders. No reminder service is implied or enabled now.
Saved / Review Later	Expose clearly and distinguish their purposes if both remain. Review Later should be an active learning queue, not a decorative bookmark list.
Privacy / data state	Eventually show device-only or synced status only when accurate. Do not pretend an anonymous local record is recoverable across devices.

4.7 Emergency reference

The emergency area is a calm **reference**, not a live assistance service.

- Cover traffic collision, freeway breakdown, traffic stop, and bad weather **when reviewed content exists**.
- Make it reachable through the established Lane experience without turning it into a social post or quiz gate.
- Use calm, official-source-reviewed steps and deliberate call/link actions only where appropriate.
- Show a prominent boundary: **pull over safely; do not use Lane while actively driving.**
- Do not request or store live location, dispatch emergency help, promise weather data, or make medical/legal claims.

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5. Stable-ID feature inventory

5.1 Inventory conventions

Every feature below has a stable planning ID. IDs make discussion, acceptance, migration, and deferred work traceable; they are not evidence that the app currently contains those identifiers.

- **Origin:** OWNER-AGREED, LATEST OWNER DIRECTION, HISTORICAL/AUDIT, or NEW/NOT APPROVED.
- **Local/browser:** may work through a single browser progress adapter and can be testable without cloud infrastructure.

- **Real backend:** means a configured, secured remote service is needed to make the claim honestly across devices/users.
- **Done check:** a concrete future demonstration, not a documentation checkbox.

5.2 Shell, navigation, and Home

ID	Origin / status not claimed	Visible frontend behavior	Required data or action	Lo ba
LN-SHELL-01	OWNER-AGREED; current shell not reverified	Lane logo at upper-left, clear route context, no odd central header text.	Logo asset and accessible label/link.	Lo
LN-NAV-01	LATEST OWNER DIRECTION; Feed gap audit-reported	Persistent bottom bar on Home, Feed, Practice, Progress, Profile.	Route state and active-tab semantics.	Lo
LN-NAV-02	OWNER-AGREED	Back arrow/Exit appears only on drilled-in detail, settings, or simulation state.	Return target and prior feed position.	Lo
LN-HOME-01	LATEST OWNER DIRECTION; unverified fix	Compact crossbar with streak on the left, lessons completed on the right, and subtle XP below.	Streak, completed count, total XP.	Lo op
LN-HOME-02	OWNER-AGREED; only partial audit evidence	Continue Learning opens actual next unfinished/current/due item.	First-session flag, item completion, active review queue, timestamps.	Lo
LN-HOME-03	OWNER-AGREED; audit snapshot only	Daily Goal sits below Continue Learning and counts useful work.	Goal target, daily qualifying-event totals.	Lo
LN-HOME-04	OWNER-AGREED retained	Leaderboard remains below Daily Goal with honest device/demo	Local fixture/device progress or future ranked service.	Lo po rar ba

		wording where appropriate.		
LN-HOME-05	LATEST OWNER DIRECTION	Remove prominent Home Saved button and redundant “next move” filler while preserving Saved access elsewhere.	Existing saved state/location.	Lo

5.3 Content, direct media, and feed cards

ID	Origin / status not claimed	Visible frontend behavior	Required data or action	Lo ba
LN-CONTENT-01	OWNER-AGREED; current runtime unverified	Mixed feed: videos, quizzes, flashcards, glossary, permit posts, visuals, 2D activities, selective 3D.	Typed content records with stable IDs and sequence/routing fields.	Lo firs
LN-VIDEO-01	OWNER-AGREED; playback unverified	19 numbered lesson videos play as direct local HTML video, not Drive/YouTube iframe.	`lessonId`, local path, poster, media metadata.	Lo firs
LN-VIDEO-02	OWNER-AGREED workflow; not proof complete	Posters/fallback and intentional sound/mute behavior.	Poster URL, audio/caption status, error state.	Lo
LN-VIDEO-03	OWNER-AGREED; audit gap	Real video `ended` event triggers its related check, not a timer or next clip.	Watch state, key-segment timestamp, linked quiz ID.	Lo
LN-CONTENT-02	OWNER-AGREED workflow	Each learner-facing item can disclose handbook source, credit/license, caption status, and approval status.	Content provenance metadata.	Lo ed

LN-CONTENT-03	OWNER-AGREED	First learner begins Lesson 1 then Lesson 2 in order; returning learner begins with relevant direct video/refresher.	First-session completion and learning selection rule.	Lo
LN-CONTENT-04	Agreed content types; label wording is a recommendation	Non-video card labels identify “Permit rule,” “Flashcard,” “Glossary,” or “Visual reference.”	Content type label.	Lo
LN-LAUNCH-01	OWNER-AGREED opening flow	A normal new/returning session opens on the selected video Feed; Home is a separate navigable destination.	Session routing plus fresh/returning selection.	Lo
LN-FEED-01	LATEST OWNER DIRECTION	Media-dominant action rail: heart, comment affordance, share, save, separate sound.	Local preference state; native share capability.	Lo int
LN-FEED-02	OWNER-AGREED safety boundary	Comment icon is omitted or has honest unavailable state until community safety requirements are met.	Feature flag/availability text only.	Pu rec
LN-FEED-03	OWNER-AGREED	Share uses native share/copy of a valid lesson link when available.	Canonical route/link and feature detection.	Lo
LN-FEED-04	Decision requires clarification	Save maps to Review Later or remains a clearly distinct Saved library.	Saved IDs and/or review queue IDs.	Lo

5.4 Quizzes, correction, Review Later, and daily review

ID	Origin / status not claimed	Visible frontend behavior	Required data or action	Lo ba
LN-QUIZ-01	OWNER-AGREED; audit reports gaps	One focused, source-linked check follows the related item at an intentional moment.	`quizId`, `topicId`, trigger rule, source/reason label.	Lo
LN-QUIZ-02	OWNER-AGREED	Question types can include multiple choice, fill blank, matching, right-of-way, hazard tap, signs, following distance, and decision pauses.	Approved question content/type contracts.	Lo
LN-QUIZ-03	OWNER-AGREED quality method	Believable, balanced choices; rotated positions; no giveaway wording; handbook citation in data.	Approved answer bank and source reference.	Lo wo
LN-QUIZ-04	OWNER-AGREED; current silent dismissal audit-reported	Important check offers answer now, key segment, Review Later, and contextual help rather than silent close.	Skip status, key-segment action, queue write.	Lo
LN-QUIZ-05	OWNER-AGREED	“I already know this” asks a harder confirmation; correct moves on, incorrect enters normal support.	Related harder question and event sequence.	Lo
LN-QUIZ-06	OWNER-AGREED	Wrong/low evidence shows calm explanation then similar—not identical—follow-up; second miss offers key segment or short activity.	Attempt history, explanation, alternate item, help/activity link.	Lo

LN-REVIEW-01	OWNER-AGREED; not proven built	Review Later is a separate active learning queue, not merely Saved.	`learnerId`, item/topic IDs, reason, queued/due/resolved timestamps.	Lo sy
LN-REVIEW-02	OWNER-AGREED	Daily in-app review draws from today, one day ago, two days ago, skipped, missed, and weak items.	Event dates, review priority, active queue.	Lo
LN-REVIEW-03	OWNER-AGREED	In-app review comes before notifications; any reminder is opt-in future work.	Preference only if reminders are added.	No for ne ba

5.5 Practice, Progress, Profile, and motivation

ID	Origin / status not claimed	Visible frontend behavior	Required data or action	Lo ba
LN-PRACTICE-01	LATEST OWNER DIRECTION; audit snapshot only	Preserve liked Practice page and reported Practice/46 structure; correct only tangible defects.	Practice questions/results in current app data.	Lo
LN-PRACTICE-02	DOCUMENTED	Future practice results may provide topic evidence, not automatic certification.	Distinct concept IDs, attempt/answer data.	Lo
LN-PROGRESS-01	OWNER-AGREED	Retain Road Ahead and useful progress; reduce overly Duolingo-like styling without structural replacement.	Current progress/roadmap data.	Lo
LN-PROGRESS-02	OWNER-AGREED safety boundary	Separate written permit knowledge from practical decision practice.	Separate knowledge and scenario evidence.	Lo

LN-PROFILE-01	OWNER-AGREED + latest request	Editable display name beside simple avatar/profile.	Same learner record field; validation.	Lo
LN-PROFILE-02	OWNER-AGREED	One anonymous learner record unites profile, avatar, XP, streak, goals, likes/saves/review state, and permit date.	Stable private learner identity and record adapter.	Lo lin
LN-AVATAR-01	Audit-reported local/demo; preserve not expand	Keep simple avatar selection and safe existing behavior.	Avatar ID, ownership only if current model uses it.	Lo
LN-XP-01	Owner direction on display; event behavior required	XP is visually quiet under crossbar and awarded only once per qualifying completion.	Idempotent event/reward key.	Lo val pu
LN-STREAK-01	OWNER-AGREED tone	Streak encourages a new session without shame or loss of historical XP.	Local calendar-qualified activity dates.	Lo
LN-GOAL-01	OWNER-AGREED	Daily Goal counts useful completed learning actions, not scrolling/replay.	Daily action ledger/count.	Lo
LN-LEADER-01	OWNER-AGREED placement; public version off	Device/demo leaderboard can remain visible, never inventing global users/rank.	Local fixture/self row.	Lo rar ba
LN-PERMIT-DATE-01	OWNER-AGREED; unverified build	Optional date changes content balance in four windows, not mastery/XP/pass probability.	`permitTestDate` setting and date mix rule.	Lo

5.6 Simulations and activity integration

ID	Origin / status not claimed	Visible frontend behavior	Required data or action	Lo ba
LN-SIM-01	OWNER-AGREED catalog; packages unverified	Related quiz offers relevant optional/recommended simulation, never unrelated opener.	Scenario/topic/quiz stable IDs and trigger relation.	Lo
LN-SIM-02	LATEST owner correction / owner-agreed	Intentional intro/loading with title, objective, controls, Start, progress/fallback—never blank screen.	Package UI and load state.	Pa ba
LN-SIM-03	Owner wants a countdown and active slow motion; durations are catalog drafts	Active slow-motion scene with synchronized countdown; no frozen traffic or auto-choice.	Scenario timing, hidden-tab handling, approved duration/mode.	Lo
LN-SIM-04	OWNER-AGREED	Specific comparable driving choices in California/US terms.	Human-reviewed choice/outcome content.	Lo pa rev
LN-SIM-05	OWNER-AGREED	Safe choice says Safe choice/Continue; unsafe/timeout show calm teaching with Retry and Review Later/Back.	Structured outcome state.	Lo
LN-SIM-06	Latest owner correction	Unsafe outcomes can show non-graphic cartoon impact/warning burst, near miss, blocked path, or braking conflict.	Visual assets/animation and outcome copy.	Lo
LN-SIM-07	OWNER-AGREED	Exit is distinct from wrong/timeout, restores feed	`onExit`, feed position, review upsert.	Lo

		position, and queues review.		
LN-SIM-08	DOCUMENTED integration contract	Portable `startScenario(container, { onComplete, onRetry, onExit })` contract yields structured outcome.	Scenario result schema and cleanup.	Lo
LN-SIM-09	DECISION NEEDED	Resolve SIM-08 built-in yellow-light versus package version, timer/default details, ID alignment, Babylon/version strategy, ZIP tracking.	Owner choice and integration map.	No rec

5.7 Content administration, community, emergency, privacy, and platform

ID	Origin / status not claimed	Visible frontend behavior	Required data or action	Lo ba
LN-EDITOR-01	OWNER-AGREED future workflow	Separate author/editor surface can add or export educational records; never becomes the learner feed.	Typed record, validation, import/export.	Lo rer
LN-EDITOR-02	OWNER-AGREED workflow	Content record includes title, topic, handbook source, direct asset path, captions, credit/license, quiz, optional activity/simulation.	Provenance/approval schema.	Lo
LN-POST-01	OWNER-AGREED	Authored permit-feed post is a source-backed Lane lesson card.	Editorial content record.	Lo

LN-POST-02	DOCUMENTED future	Editorial blog/content record remains an authoring concept separate from comments.	Exportable/editorial metadata.	Lo op
LN-COMMUNITY-01	Future/off; comment icon is UI direction only	Public user comments or local-hazard discussion appear only after moderation/privacy design.	Authorship, reports, moderation state, rate limit, policies.	Re ba op
LN-EMERGENCY-01	OWNER-AGREED; current screen unverified	Calm reference for collision, breakdown, traffic stop, bad weather when content exists.	Reviewed local content and safe links.	Lo
LN-LEGAL-01	Final release work	Privacy/TOS links and notices match actual fields/features.	Data inventory and owner/legal review.	Ma mu ba
LN-PWA-01	Audit-reported app shell only	Offline app shell and local progress survive an offline revisit after prior online load.	Service worker/cache strategy.	Br
LN-PWA-02	Separate media reliability gate; full offline video is deferred	Exclude large videos from automatic precache. Online direct playback works; offline unavailability has a useful fallback.	Explicit service-worker exclusions, media range/codec behavior, offline notice/poster.	Br ho
LN-BACKEND-01	DOCUMENTED / unverified configuration	One adapter mediates learner/progress reads/writes; UI does not call future Supabase directly.	Repository adapter and typed payloads.	Lo lat
LN-BACKEND-02	Planned only	Remote foundation supports learner/profile, video progress, events,	Configured Supabase, migrations, RLS,	Re rec

		mastery, Review Later—not community/leaderboard/editor.	anonymous auth/session.	
LN-SECURITY-01	OWNER-AGREED boundary	Browser uses only publishable client configuration; no secrets exposed.	Environment validation and server-only secrets.	Re ad

5.8 Inventory notes that prevent scope mistakes

- **Local behavior is still real behavior**, but it must be described as device/browser-local until sync is proven.
- **Anonymous does not mean insecure public string identity.** Do not use a guessed or shared `demo-learner` identifier as an authorization boundary. See Section 8.
- **A visual comment button is not a community backend.** The public social feature remains outside the proposed backend foundation v1.
- **A source manifest is not a CMS.** It is a local content workflow and can be valuable before any remote administration exists.
- **A video file is not its metadata.** Store video bytes/paths separately from the content record that explains its title, topic, source, captions, credit/license, and linked learning items.
- **A demo leaderboard is not a public leaderboard.** A future public system needs authenticated/linked identity, privacy, server-validated XP, anti-cheat rules, and moderation decisions.

5.9 Personalization, evidence, and content stewardship

ID	Origin / status not claimed	Frontend effect	Required data/action	Local vs backend
LN-EVIDENCE-01	Owner-requested signals; tracking unverified	Videos/questions respond to what was watched, attempted, answered, retried, or deferred.	Answer correctness, submissions, help/retry/activity events, active time, genuinely watched intervals/key segment.	Local first; cloud sync optional.

LN-ALGO-01	Four-signal design agreed; exact missing-evidence policy unresolved	Proceed, reinforce, explain, replay key segment, or offer practice through one tested module.	Accuracy, Work, Attention, Time-quality inputs and unrounded master score.	Pure local calculation; remote validation optional.
LN-ALGO-02	Documented review/smoothing method	Weak, skipped, stale, and safety-important topics return without erasing prior evidence.	Prior mastery, new session, last practiced, skip count, topic safety importance.	Local first.
LN-READINESS-01	Owner requested confidence; prediction not validated	Show limited/developing/strong in-app knowledge and decision-practice evidence, with gaps to review.	Distinct topic/concept coverage and question/scenario outcomes, not XP alone.	Local first.
LN-CONTENT-QA-01	Original owner request; final completion unverified	Useful captions/descriptions, balanced quiz choices, accurate California rule context.	Caption/script/video checks against the handbook; corrections and reviewed status.	Editorial/local workflow, not a new backend dependency.
LN-IMPACT-01	Original owner request; not a runtime service	Short, factual impact/context may accompany relevant lessons without sensationalism.	Source-cited Novato-area evidence where relevant; date/location and uncertainty recorded.	Static editorial data.
LN-LIBRARY-GROWTH-01	Owner requested more useful material	Add reviewed videos/posts/visuals when they fill real coverage	New-content review, source/license, topic mapping	Local/static initially; future editorial service optional.

		gaps; keep current 19 IDs/order stable.	and linked questions.	

6. Simulation catalog and ownership

6.1 Exact locked catalog

The current catalog target is exactly **10 scenarios**: **7 animated 2D** scenes and **3 driver-seat 3D** scenes. Each occurs after its related quiz, not as a feed opener.
The 3–5 second values below are **earlier catalog draft settings**, not independently approved/tested final timings. Keep the owner-requested countdown, but validate legibility, pacing, and any accessibility/practice mode before finalizing durations. The scenario count/types and latest active-slow-motion correction are separate decisions.

ID	Type	Scenario and learner decision	Safe outcome	Unsafe / timeout teaching outcome	Timer
SIM-01	2D	**Four-Way Stop: Who Goes First?** After complete stops and simultaneous arrival, yield to the vehicle on the right; separately yield to pedestrians crossing.	Relevant road users clear; learner proceeds when safe.	Conflict warning before contact; explain the specific tie/yield rule.	4s draft

SIM-02	2D	**Unprotected Left Turn: Wait for Oncoming Traffic.** Yield to close oncoming traffic and pedestrians.	Wait, then turn when clear.	Oncoming conflict warning; explain yield duty.	4s
SIM-03	2D	**Roundabout Entry: Yield Before Joining.** Wait for safe gap, then enter right.	Safe entry after circulating traffic gap.	Conflict warning; explain yield to circulating traffic.	4s
SIM-04	2D	**Lane Ends: Merge Early and Deliberately.** Signal, mirrors, shoulder check, merge into gap.	Safe prepared merge.	Lane-end conflict warning; explain planning and space.	5s
SIM-05	2D	**Blind-Spot Check: Motorcycle Beside You.** Shoulder-check and wait.	Motorcycle revealed; learner creates space.	Motorcycle highlight; explain mirror blind spots.	4s
SIM-06	2D	**Emergency Vehicle: Clear the Intersection, Then Pull Right.** Finish clearing, then move right/stop safely.	Emergency vehicle passes with access.	Blocked-access warning; explain no mid-intersection stop.	4s
SIM-07	2D	**Railroad Tracks: Do Not Enter Without a Clear Exit.**	Remain safely before tracks.	Trapped-on-tracks warning; no graphic crash;	5s

		Stop before tracks until fully clear.		handbook recheck remains required.	
SIM-08	3D	**Yellow Light: Stop if You Can Do So Safely.** Brake smoothly before the limit line.	Clean safe stop.	Red-light conflict warning before collision.	3s
SIM-09	3D	**Freeway Entrance: Match Traffic and Merge Into a Safe Gap.** Signal, match appropriate speed, shoulder-check, merge.	Smooth merge into available gap.	Too-close entry warning; explain freeway traffic right-of-way.	5s
SIM-10	3D	**Right Turn Scan: Pedestrian and Bicyclist at the Corner.** Wait, scan, yield, turn when clear.	Vulnerable road users clear; safe turn.	Highlight missed road users; explain green light does not remove yield duty.	4s

6.2 Scenario lifecycle and result contract

Plain Text

```
Related quiz
   ↓
Invitation: optional if doing well / strongly recommended after relevant high-s
   ↓
Intro/loading: title + objective + control instruction + Start + progress/fallb
   ↓
Active slow-motion decision window (3–5 seconds; motion continues)
   ↓
Safe ————————> Safe choice + brief why + Continue / Back to lesson
```

```
Unsafe ———> Calm warning/teaching + Try again / Review Later / Back
Timeout ———> Calm teaching + Try again / Review Later / Back
Exit ————> Clean resource disposal + return to same feed position + Review La
```

Each approved portable package should be able to start through:

Plain Text

```
startScenario(container, { onComplete, onRetry, onExit })
```

The integration result must distinguish outcomes rather than flattening everything into `success`:

Plain Text

```
{
  scenarioId: string,
  topicId: string,
  outcome: "safe" | "unsafe" | "timeout" | "exit",
  selectedAction?: string,
  elapsedMs?: number,
  retryCount: number,
  handbookSource: string
}
```

6.3 Ownership: GPT/Codex versus Claude/Claude Code

Owner	Owns	Must not do
GPT/Codex / simulation workspace	Package-level scene design, meaningful choices, loading/Start state, active slow-motion timing, safe/unsafe/timeout/exit UI, cartoon warning symbols, asset manifest, cleanup, standalone fallback.	Redesign Lane Home, Feed, navigation, profile, progress, direct-video integration, or backend architecture.
Claude / Claude Code / Lane integration	Existing Lane shell, routes, persistent navigation, feed ordering, direct local videos, non-simulation UI, learner progress seam, Review Later linkage, future backend adapter, package import.	Redesign an approved simulation package or invent new scenario behavior without owner request.

Owner / content reviewer	Final handbook correctness, choice/outcome approval, package/replacement decisions, content sources/licenses, and unresolved policy choices.	Treat a draft ZIP manifest or old audit claim as final approval.

6.4 Integration gates still requiring explicit resolution

- The supplied ZIP reportedly differs from the agreed target on 2D camera style and timer defaults; package behavior was not run.
- The audited built-in yellow-light simulation reportedly has two choices, while catalog **SIM-08** has three and a 3-second timer. Keep/replace is a decision gate.
- All package manifests reportedly need human handbook review.
- Shared lesson/topic/quiz/scenario IDs need an authoritative registry.
- Babylon/version compatibility, ZIP storage/tracking, and package import strategy need a technical decision before 3D import.
- Simulation result storage and Review Later support must exist through the normal progress seam before Lane claims end-to-end simulation learning.

---

7. Explainable learning and readiness

7.1 The four learning signals

Lane’ s recommended personalization is intentionally simple enough to explain. It must optimize useful learning—not likes, popularity, follow count, or total time spent.

Signal	Plain-English question	Examples of evidence	Guardrail
Accuracy	Did the learner choose correctly, learn from correction, and remember later?	First answer, correction answer, later review.	Watching/trying cannot substitute for knowing.
Work	Did the learner engage with available learning actions?	Submitted answer, retry, help, activity completion.	Missing/unassigned work must not be silently treated as personal failure.

Attention	Did the learner reach the important instructional material?	Unique/max watch progress, key segment reached, purposeful interaction.	Screen-open time alone is not attention.
Time quality	Was the answer at a healthy pace rather than an instant guess or very long stall?	Active answer time versus expected range.	Small diagnostic only —not speed scoring, punishment, or a timer game.
The documented formulas, routing bands, review priority, and smoothing logic belong in dedicated algorithm materials. The frontend should explain **outcomes**, not display algebra on every card:			

- “You watched the key part; try one similar question.”
- “This rule is back because it was skipped.”
- “This is a short refresher for a safety topic.”
- “More evidence will help us suggest what to practice next.”

7.2 Mastery, review priority, and readiness are distinct

Concept	What it answers	Frontend implication
Current-session mastery	How well did the learner demonstrate this topic now?	Proceed, reinforce, correct, show key segment, or suggest practice.
Topic smoothing	How should a new session update prior evidence without overreacting to one lucky guess/bad session?	Progress changes gradually; do not claim AI prediction or permanent mastery.
Review priority	Which topic should return next?	Prefer weak, old, skipped/Review Later, and safety-important topics.
Written knowledge evidence	Does the learner show enough distinct permit-rule knowledge?	“Limited / developing / strong evidence,” never a pass

		guarantee.
Practical decision-practice evidence	How did the learner handle scenarios/activities in-app?	Separate from written knowledge; never a real-world certification.
The documented correction that an example raw score of **0.59875** is below the 0.60 routing threshold must be preserved. Display rounding may show 0.60, but it must not silently change the support route.		

7.3 Work-score normalization conflict — decision needed

> **Do not automatically alter this.** The supplied written method describes fixed Work weights with unavailable correction/activity signals contributing zero. The supplied reference `algorithm.ts` normalizes over the signals that are actually available. A previous assistant's missing-signal rule change was not owner-approved.
> | Decision to make | Option A | Option B | Required before implementation |
> |---|---|---|---|
> | **How unavailable Work evidence is handled** | Fixed weights: unavailable signals contribute zero. | Eligibility-aware normalization: unavailable/unassigned signals are excluded from the denominator. | Owner selects a policy, the reason is documented in plain English, and tests show outcomes for assigned vs unavailable work. |
> Until this is resolved:

- do not claim a final Work formula;
- do not silently rewrite the reference implementation;
- do not tell learners they “did not try” merely because an activity was unavailable; and
- preserve raw evidence so a later policy can be explained and tested.

7.3a Formula reference — proposed heuristics, not trained coefficients

The user asked for four simple formulas and a master formula. These are the existing proposed values, included so this master is understandable on its own. They are design starting points, **not experimentally validated pass predictors**.

Plain Text

```
A = .60*C1 + .25*C2 + .15*C3
    C1: first-answer correctness; C2: correction-answer correctness;
    C3: later-review correctness/rate (historical neutral prior .50 if unknown)

W numerator = .40*Q + .30*R + .20*I + .10*H
    Q: submitted; R: retry completed; I: assigned activity completed;
    H: opened explanation/help.
    Denominator / unavailable-signal policy: unresolved as described above.

N = .50*V + .30*K + .20*E
    V: genuinely watched fraction; K: key segment actually watched;
    E: purposeful engagement where available, with an explicit missing-data rul

r = active answer seconds / expected question seconds
T = 1.00 for .50 <= r <= 2.00
    .70 for .25 <= r < .50
    .45 for r < .25
    .75 for 2.00 < r <= 4.00
    .55 for r > 4.00
    Historical expected time: 25 seconds; tune per question, not every scenario

M = .55*A + .15*W + .20*N + .10*T
    >= .80: strong evidence; >= .60: developing;
    >= .40: needs support; below .40: instruction/support needed.

Updated topic mastery = .70*previous mastery + .30*session M
P = .50*(1-current mastery)
  + .20*min(days since practice/7, 1)
  + .20*min(skips/3, 1)
  + .10*topic safety importance
    Higher P suggests earlier review, not a higher learner grade.
```

Additional consistency checks: the reference code supplies correction credit when the first answer was correct and no correction was assigned, while historical examples may treat that missing signal differently. Its attention default also infers engagement from video completion when no explicit engagement ratio is supplied. Reconcile these cases before claiming a finalized scoring method; never describe inferred/default evidence as an observed learner action. A help tap, long dwell time, or completed video is not proof of comprehension. For a first-ever topic, choose an explicit initialization rule rather than inventing previous evidence.

Recommendation, not an approved change: normalize only genuinely applicable opportunities; retain unavailable evidence as unavailable; let correct knowledge and later recall outweigh passive watching. Compare this approach against the historical method in

tests before selecting it. This is an explainable rule-based learner model: it updates each person's evidence, but does not train the weights automatically.

7.4 Optional permit-test date

The learner may optionally set a permit-test date. It changes **content balance**, not the learning formula or the promise of a passing result.

Time until test	More likely content mix
No date or more than 30 days	Practical driving videos, foundations, occasional flashcards/glossary/visuals.
14–30 days	More handbook posts, signs/rules, and short quizzes mixed into lessons.
7–13 days	More weak-topic refreshers, permit-style posts, signs, numbers, definitions, quizzes.
0–6 days	Calm short review sessions: missed rules, Review Later, flashcards, quick questions.
No date state should be fully useful. Clearing the date returns to ordinary mix. A countdown must not shame, pressure, or imply a passing guarantee.	

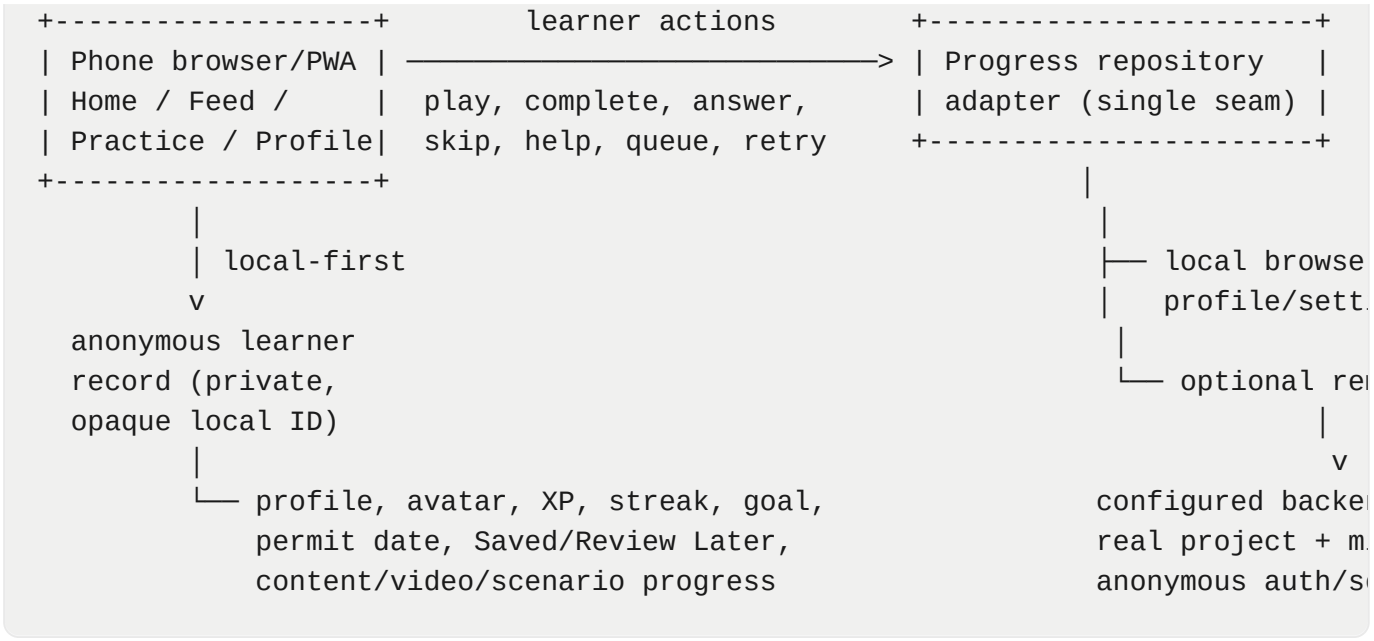
---

8. Learner/content/progress/backend model

8.1 Simple text diagram

Plain Text

```
                    EDITORIAL / CONTENT WORKFLOW
      original local media files ──────┐      handbook/source review
      Drive/archive source storage ────┼──>   captions / credit / license / ap
      posters, glossary, artifacts ────┘                 |
                                                         v
                                            Content manifest / records
                                            contentId, lessonId, topicId,
                                            direct asset path, quizId,
                                            simulationId, provenance
                                                         |
                                                         v
```



8.2 Shared stable identifiers

Entity	Minimum stable identifier	Purpose
Learner	`learnerId`	Private key joining the local learner's profile, settings, progress, and events.
Content item	`contentId`	One record for a video, permit post, flashcard, glossary, visual, or activity card.
Lesson/video	`lessonId`	Ordered lesson identity independent of filename/path.
Topic	`topicId`	Shared concept link across questions, videos, practice, review, and scenarios.
Quiz/question	`quizId` / `questionId`	Source-backed question and its variant/attempt history.
Scenario	`scenarioId`	SIM-01 through SIM-10 package/result identity.
Learning event	`eventId` / idempotency key	Prevent duplicate rewards/progress writes after retry, refresh, or replay.

Review item	`reviewKey`	One active learner + content/topic queue item, upserted/reactivated rather than duplicated.

8.3 Anonymous identity without insecure public identity

A local-first learner can be anonymous to Lane without using a weak shared identity.

- Generate/store an **opaque, per-browser learner ID** through the one progress adapter; do not expose it in UI, URLs, leaderboard rows, or public payloads.
- Do not treat a hard-coded string such as `demo-learner`, a display name, or a client-provided arbitrary user ID as proof of ownership.
- If a remote backend is approved later, use its configured anonymous-auth/session mechanism or other server-issued identity. The server/RLS layer must decide which row a session may read/write.
- Browser clearing, private browsing, device changes, and unlinked anonymous sessions can lose recoverability. Tell learners the truth rather than promising backup.
- Future sign-in should **link** the existing anonymous learner record and avatar/progress rather than reset the learner.

8.4 Local-first versus real backend boundaries

Capability	Valid local/browser first?	Requires a real configured backend to claim honestly?	Why
Display name, avatar, XP, streak, daily goal	Yes	Only for cross-device sync/recovery.	One local learner record can support it.
Continue Learning, first/returning routing	Yes	No, unless cross-device continuity is promised.	Uses local completion/review evidence.
Review Later and daily review	Yes	Only for sync across devices.	Queue/event logic works locally.
Video watch progress and scenario results	Yes	Only for sync/audit across devices.	Stable IDs plus local event store are sufficient initially.
Direct local video playback	Yes	No.	Media files and content manifest are

			local/static concerns.
Content metadata/editor export	Yes, as manifest/import/export.	Yes for multi-user remote editorial workflow.	Content bytes/metadata can be local initially.
Public comments or local-hazard discussions	No public version.	Yes.	Needs authorship, reports, moderation, rate limits, privacy, and policies.
Real leaderboard	Device/demo only.	Yes.	Needs linked identities, server-validated XP, privacy, moderation, anti-cheat.
Push reminders	In-app review yes.	Usually yes.	Needs opt-in, service/device delivery, and privacy management.

8.5 Proposed remote foundation boundary

If and only if discovery confirms an approved Supabase project and the owner wants remote persistence, keep v1 narrow:

Plain Text

```
Included v1:
  learner profile/settings
  video progress
  append-only learning events
  topic mastery/readiness evidence
  Review Later queue
  explainable next-item/review selection
Explicitly not v1:
  public comments/posts
  public likes
  social graph / DMs
  real leaderboard
  full admin CMS/editor
  live emergency dispatch/weather service
```

Server safety requirements

- Browser configuration may use only a project URL and publishable client key when a real project exists.
- Never place a service/secret key in browser code, a `VITE_*` variable, source control, screenshots, or a prompt.
- Validate stable IDs, enum/range values, and event idempotency on the trusted side when remote writes exist.
- Use own-row RLS/policies for learner-owned records; prove another session cannot read/write them.
- Preserve a local action when remote sync fails, show a retryable honest state, and avoid double-counting on retry.
- Do not apply migrations, guess project details, or claim remote connectivity without a configured project and an end-to-end anonymous save/read test.

---

9. New friendly recommendations — not yet approved

These recommendations are deliberately small. They do not replace agreed features, add addictive loops, create new public social surfaces, or settle unresolved algorithm/backend decisions.

Priority	Proposal ID	Recommendation	Learner benefit	Dependency	Why it remains separate
P0	**NEW-RESUME-01**	Add a small **Saved resume point**: "Continue at 1:24" or "Resume the key part," only after actual progress is recorded.	Reduces friction when returning to a real video without creating a feed loop.	Local video progress; stable lesson ID.	Helpful refinement, not explicitly owner-approved.
P0	**NEW-WHY-REVIEW-01**	Add one plain reason label on every returning item: "Skipped," "Due for	Makes personalization transparent and reduces the feeling of random interruption.	Review reason/priority data.	Aligns with agreed explainability but exact UI i new.

		review," "Weak evidence," or "Safety topic."			
P0	**NEW-ACCESS-01**	Add a unified **captions, text alternative, and reduced-motion** affordance/status. If captions are pending, say so plainly and offer text key points where reviewed.	Supports hearing, attention, motion, and noisy-environment needs.	Content caption/text status; frontend preferences.	Caption completion and exact UI are not yet approved/verified.
P1	**NEW-TIMER-01**	Offer **Calm practice** and separate **Challenge timer** modes for appropriate activities, with remembered local preference. Required catalog timers remain an explicit integration decision.	Lets learners practice without making all learning a reaction-speed test.	Simulation/activity design and accessibility review.	Must not override the locked scenario contract or unresolved timer-default decision.
P1	**NEW-SYNC-01**	Show an **honest sync state** only when relevant: "Saved on this device" or "Sync will retry."	Prevents accidental belief that anonymous browser data is cloud-backed.	Adapter state; future remote sync if added.	Wording and technical trigger need owner approval.

P1	**NEW-REPORT-01**	Add a small **Report content issue** action for a learner-facing source/caption/problem report—not a public discussion channel.	Lets teens flag an incorrect sign, broken video, or caption problem safely.	Local mail/form/export workflow initially; moderation workflow later.	New support process; mus not imply live moderation/r sponse.
P2	**NEW-SUMMARY-01**	End each intentional session with a tiny optional summary: what was completed, one rule retained, and one Review Later item if any.	Gives closure without badges, social comparison, or pressure to keep scrolling.	Local event/review state.	Exact content and visual treatment are new.
P1	**NEW-FIND-01**	A simple topic/glossary finder: search “stop signs” or filter Saved/Review Later by topic.	Find a useful rule without scrolling the whole feed.	Local content index initially; no paid search service needed for this small library.	Search was invoked as a skill earlier, not approved as a shipped UI feature.
P2	**NEW-GOAL-01**	Let the learner choose a small realistic daily goal, such as a short lesson or a few checks, with clear qualifying actions.	Gamification feels achievable and under the learner's control.	Local settings and existing daily-goal logic.	Choice of targets/rewar rules require a small explicit desig decision.
P2	**NEW-UNDO-01**	Brief Undo after removing a saved/review	Prevent accidental loss without constant	Local action history; same progress seam.	New interaction detail; does not excuse

		item or changing a non-destructive preference.	confirmation popups.		unsafe data deletion or bypass moderation.

Recommendation guardrails

- No autoplay loop designed to maximize session duration.
- No reward chest, loot system, social streak pressure, random prize, or avatar economy expansion.
- No public follower counts, DMs, leaderboards with invented peers, or fake engagement.
- No “You are ready to drive” outcome from in-app evidence.
- No notification without an explicit opt-in and a future service/privacy design.

10. Lean build iterations and deferred registry

10.1 Small slices: build the learning loop before breadth

Iteration	Small slice	Acceptance criteria before next slice	Explicitly preserve/defer
0 — Discovery	Inspect the actual VS Code project, routes, storage, assets, design tokens, current tests, service worker, package versions, and feature differences.	System/feature/asset map and discrepancy list distinguish current code from this plan. No code change is called complete without command evidence.	Do not assume old starter backend, audit snapshot, assets, packages, or Supabase configuration still match.
1 — Shell correction	Logo/header cleanup, Home order, persistent Feed navigation, safe-area/focus/touch fixes.	Home and core tabs pass visual/keyboard checks at 360×800, 390×844, 768×1024, 1366×768; no hidden last action.	Keep Practice, Road Ahead, Profile, existing routes and design language.

2 — One real lesson vertical slice	One owner-approved local video record: direct playback, poster/fallback, actual `ended`, linked focused quiz, local progress write.	Play/load/error/mute/end/quiz/reload work on desktop plus one physical iPhone and Android; provenance fields show status.	Do not claim all 19 media files, caption completeness, or offline video yet.
3 — Learning choice slice	Explicit skip/key-segment/help/Review Later and supportive correction for that one lesson.	No silent close/scroll escape; queue dedupes; wrong route uses a similar question; exit does not mark mastery.	Do not introduce random checks, scroll lock, notifications, or a public save/social feature.
4 — Home truthfulness	Continue Learning selects fresh/next/due local item; crossbar/goal/XP idempotency; profile display name.	Fresh, returning, due-review, duplicate-submit, refresh, replay, and name-reload fixtures pass.	Public leaderboard, real sync, account migration, and formula-policy changes remain deferred.
5 — Content library expansion	Map all 19 assets and add approved mixed card types through stable content/topic IDs.	Every item has required metadata/status; lesson order is preserved; content mix is intentional; no filler.	Content CMS, mass uploads, and public release claims remain deferred.
6 — Simulation integration	Import one approved package only after manifest/ID/version/load/result checks; connect to quiz and Review Later.	Start/loading, active slow motion, safe/unsafe/timeout/exit, cleanup, result persistence, and feed-position return pass.	Other nine packages, timer defaults, SIM-08 replacement, and package architecture stay in registry until approved.
7 — Explainable progress	Add review reason labels, separate knowledge/practical evidence, permit-date mix after Work policy decision.	Deterministic algorithm/date/priority tests pass; UI makes no prediction/certification claim.	No hidden reweighting, popularity ranking, or irreversible readiness label.
8 — Optional remote foundation	Only if owner confirms and a real configured project exists: adapter, anonymous service session, narrow	Remote anonymous save/read plus own-row denial, idempotent retry, offline fallback, and	Community, public likes, public ranking, full editor, and reminders remain outside v1.

	schema, RLS, local fallback.	build/walkthrough evidence pass.	
9 — Release readiness	Physical-device/PWA/media/caption/content/privacy/accessibility review.	Evidence covers actual enabled features, device media behavior, source/caption/handbook workflow, legal links, and no-use-while-driving.	No claim of award, zero bugs, professional legal approval, or public community unless separately demonstrated.

10.2 Deferred registry — preserve, do not silently drop

Deferred ID	Feature or decision	Reason deferred / trigger to revisit
DEF-01	Public comments and local hazard/permit-tip discussion	Requires moderation, reporting, ownership, privacy/minor rules, rate limits, RLS, and operating process.
DEF-02	Real leaderboard	Requires linked identity, nickname-only display, server-validated XP, anti-cheat, privacy, moderation, ranking/tie policy.
DEF-03	Account sign-in and cross-device recovery	Add only after anonymous-local learning flow is stable and migration/linking is designed.
DEF-04	Remote Supabase foundation	Requires real project/configuration, approved migration/RLS design, adapter, and live end-to-end test.
DEF-05	Full content CMS/upload service	Local manifest/export proves workflow first; remote collaborative editor is a later separate scope.
DEF-06	Push reminders	Requires learner opt-in, delivery service, privacy policy alignment, and quiet notification design.

DEF-07	All-video offline caching	Requires media size, service-worker Range behavior, storage, codec, and physical-device strategy.
DEF-08	Final Privacy Policy / Terms	Must reflect actual release features/data after owner/legal review; a placeholder is not completion.
DEF-09	Work-score normalization policy	Owner must choose fixed-zero versus eligibility-aware normalization with test evidence.
DEF-10	SIM-08 and package compatibility decisions	Resolve existing yellow-light versus package, camera/timer mismatch, IDs, Babylon version, ZIP storage/tracking.
DEF-11	Final caption/license/credit/handbook/Novato review	Editorial evidence must be checked per item; not a runtime backend dependency.

11. Learner journeys and verification plan

11.1 Essential learner journeys

Journey	Expected experience	Evidence to capture
Fresh learner	Opens directly to Video Lesson 1; Lesson 2 remains in intended order; small supporting cards do not overwhelm.	Initial local record, selected content IDs, rendered order, navigation state.
Returning learner	Home/Feed offers a relevant direct video: next unfinished, current resume, or due refresher with a plain reason.	Stored progress fixture, selection reason, opened lesson ID.
Test-near learner	Optional 0–6-day permit date shifts mix toward calm short review, missed rules, flashcards, and quick questions; no pass promise.	Boundary-date fixtures, selected item types, unchanged mastery calculation.

Early skip	Before a key segment, learner sees answer now, key segment, or Review Later; no silent loss of important rule.	Event sequence, queue state, result after reload.
Wrong answer	Learner sees specific supportive explanation, a similar question, then key segment/activity only after another miss if needed.	Copy snapshot, question IDs prove variation, no shame/lock wording.
I already know this	Harder confirmation; correct moves on efficiently, incorrect receives normal support.	Two branch results and stored events.
Review Later	Item is added once, later appears with reason, can be resolved/re-queued, and does not duplicate.	Queue upsert/reactivation tests and daily-review fixture.
Simulation safe outcome	Intentional Start, active slow-motion decision, Safe choice/Continue, normal learning record update.	Package/video capture, structured scenario result, feed return.
Simulation wrong/timeout/exit	Calm non-graphic warning; Retry and Review Later/Back are distinct; exit never implies mastery.	Each outcome result, no XP farming, exactly one active review item on exit.
Replay/refresh abuse attempt	Rewatch/retry/refresh cannot mint XP or goal/streak progress repeatedly.	Event idempotency key, XP/daily totals before/after, scenario retry count.
Device-only learner	Profile says device-only when accurate; no false promise of account recovery; app remains useful offline within tested limits.	Clear status, reload/offline state, browser-clear caveat documented.

11.2 Required verification matrix

Area	Minimum checks	Do not claim until demonstrated
Mobile web	360×800 and 390×844; portrait scroll end; safe-	Phone-ready or no-overlap.

	area/nav spacing; touch targets; keyboard where available.	
Tablet/desktop	768×1024 and 1366×768; readable centered feed; full practice/progress scroll; active nav.	Responsive completion.
Accessibility	Focus order/outline, screen-reader names, contrast measurement, reduced motion, text scaling/zoom, captions/text states, no color-only outcome.	Accessible experience or caption support.
Video	Direct playback/load failure/poster, audio/mute, actual ended event, correct quiz, codec behavior on physical iPhone + Android.	All 19 work, caption coverage, or reliable mobile playback.
PWA/offline	Online-first shell install/cache behavior; offline Home/core local screens; local action retention; deliberate video-cache behavior.	Offline real-video support from app-shell evidence.
Learning logic	Fresh/returning, skip, help, confirmation, correction, Review Later dedupe, date buckets, priority, raw threshold boundary.	Explainable algorithm or readiness claims.
Rewards	Duplicate submit, refresh, replay, retry, simulator replay, same-day activity and streak boundary.	XP/goal/streak integrity.
Simulation	Start/loading/fallback, active timer/hidden tab, correct/wrong/timeout/exit, cleanup, result contract, responsive fallback.	Integrated package completeness, performance, or safety validation.
Remote service	Configured project, anonymous session, own-row RLS allow/deny, idempotent retry, offline fallback, no browser secret.	Backend, sync, authentication, or public data claim.

Content/legal	Per-item source/license/caption/handbook/approval status; emergency review; privacy/TOS feature/data match.	Public-release, legal, or source-complete claim.

11.3 Command-evidence rule

This document makes no completion claim without evidence from the actual implementation environment. A future build task should record the commands and results it actually ran, for example:

Plain Text

```
package install / typecheck / unit tests / production build
responsive browser checks at required viewport sizes
physical iPhone and Android video checks
offline/service-worker checks
remote migration/RLS tests only after configured project approval
```

The exact commands depend on the inspected repository. Do not invent command names, test results, or a deployment status from this document or old audit material.

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12. Exclusions, unknowns, and source map

12.1 Explicit exclusions and non-claims

This master plan does **not** authorize or prove:

- a native iOS/Android app, a new framework, React, Next.js, tRPC, Drizzle, Express, or a structural rewrite;
- production Supabase configuration, a deployed API, remote migration, anonymous auth session, RLS, or cloud synchronization;
- public comments, user posts, DMs, social graph, real likes, or community moderation;
- real/global/class leaderboard, supported rankings, public learner names, or public location data;
- driver certification, DMV affiliation, permit-test pass prediction, real-world driving readiness, medical/legal advice, live emergency dispatch, or live weather;
- verified playback/caption/codec/offline support for all 19 videos;

- final source, credit, license, caption, handbook, glossary-permission, Novato/local-impact, or legal-review completion;
- completed 10-package integration, physics/performance validation, or a resolved yellow-light/package decision;
- a settled Work normalization policy; or
- an award, a bug-free release, a successful deployment, or an “already done” claim.

12.2 Open questions requiring owner or implementation discovery

Question	Why it matters	Required next evidence/decision
What does the actual latest VS Code repository contain?	Supplied audit and starter documents conflict on backend/features.	Fresh system map, run commands, asset map, feature test evidence.
Where should Saved live, and does it become Review Later?	Avoid duplicate concepts and lost data/discoverability.	Owner interaction decision; migration/UX test.
What is the exact real/local leaderboard wording?	Keep owner-requested card without fabricated public status.	Owner copy approval and local/remote state test.
Should unassigned Work signals count as zero or normalize out?	Changes learner routing fairness.	Explicit owner policy plus test cases.
Is SIM-08 built-in scene retained or replaced?	Existing scene and target catalog differ.	Owner/package decision, manifest/ID/version verification.
Which 19 assets have compatible formats, captions, and completed credits?	Direct playback and public-release accuracy depend on it.	Per-asset watch, device test, provenance record.
Is a remote backend actually desired after local-first slices?	Prevents premature service complexity and security claims.	Owner approval plus configured project discovery.
Are comments intended as an unavailable affordance or omitted until safe?	Avoid fake/dead social UI.	Owner UI decision and future moderation plan.

12.3 Source map used for this consolidation

Source	Role in this document	Confidence/use limitation
`/home/ubuntu/Lane-Agreed-Features-Source-of-Truth.md`	Primary agreed scope: product, 19-video direction, flows, simulations, algorithm boundaries, profile, review, emergency, content workflow, build boundaries.	Product intent, not current-code proof.
`/home/ubuntu/upload/pasted_content_5.txt`	Latest owner feedback on header/logo, Home hierarchy, navigation, feed controls/check placement, preserved Practice/Progress/Profile, editable name, simulation corrections.	Owner observation/direction, not completion proof.
`/home/ubuntu/lane-master-review/layout.md`	Home/nav/responsive/PWA/local persistence evidence, gaps, and safety qualification.	Supplied audit snapshot, not live re-test.
`/home/ubuntu/lane-master-review/feed.md`	Direct media, 19-video/mixed-feed, question quality, skip/help/review, social-control boundaries.	Supplied audit/document synthesis, not full asset/device proof.
`/home/ubuntu/lane-master-review/simulation.md`	Locked SIM-01–SIM-10 catalog, timing/outcomes, package contract/ownership, unresolved package gates.	Packages and runtime were not run in this task.
`/home/ubuntu/lane-master-review/algorithm.md`	Four signals, mastery/review distinction, corrected raw threshold, date mix, Work normalization conflict.	Formula/design evidence; implementation connection unverified.
`/home/ubuntu/lane-master-review/gamification.md`	Anonymous learner/profile, avatar, XP/streak/goal, leaderboard and anti-farming boundaries.	Local/demo audit evidence; not a remote service claim.
`/home/ubuntu/lane-master-review/backend.md`	Content/admin/community/emergency/privacy/backend boundaries, local-first adapter, security/RLS requirements.	Planning and audit synthesis; remote setup unverified.
Original `pasted_content.txt`, `_2.txt`, `_3.txt`, **and** `_4.txt` **under** `/home/ubuntu/upload/`	Original brief, app concepts, supplied repository/design	Distinguish the owner request from the coding agent's dated observations and assistant-

	notes, and Claude audit transcript.	written implementation instructions.
`Lane-10-Scenario-Catalog.md` and `Lane-GPT-Simulation-Refinement-Prompt.md`	Specific ten-scene catalog and later owner-feedback translation.	Timer values and packages still need testing; newest slow-motion/success-feedback direction takes precedence over older freezes.
`driver-app-learning-algorithm.md`, `driver-app-algorithm-derivation-rough.md`, and historical `shared/algorithm.ts`	Proposed formulas, arithmetic corrections, and conflicting optional-evidence implementations.	Historical reference only; do not copy inconsistent defaults or claim live integration.

12.3a How to use this single master

Give this document to Claude Code with the **actual repository**, approved logo/design assets, numbered video folder, and approved GPT simulation packages. Ask it to use stable feature IDs when reporting implementation and deferrals. It does not need the six internal area-review files to understand the intended app. Read existing `CLAUDE.md` / `AGENTS.md` first, use latest owner directions to resolve design conflicts, and keep the feature registry accurate after each small slice. This document is not an instruction to activate every future/public feature at once.

12.4 Closing recommendation

The strongest next version of Lane is not the largest feature list. It is a **truthful, teen-friendly, evidence-guided learning loop**: a stable social-feed-like shell; direct, well-described lesson media; clearly placed checks with dignified deferral; supportive Review Later; preserved Practice and Road Ahead; small, non-addictive motivation; and carefully bounded simulation and backend work.

Build it in the listed slices, keep every deferred feature visible in the registry, and let fresh repository evidence—not old documentation—determine what is actually complete.